Practice list: 50 easy problems

Fifty beginner problems that teach you Python by using it. Do them in order: each stage only needs the lessons before it.

They come from two sites. **GfG** is GeeksforGeeks; these are from its "Basic" level (the easiest it has) and its Python practice set. The LeetCode ones are the few that almost everyone solves (85% or more of attempts pass). Make a free account on both.

On either site, choose **Python3** as the language. Tick the box when you've solved one.

Stuck for 20 minutes? Look back at the lesson. Stuck for 40? Read the hints or solutions on the site, close them, and write it yourself.

Stage 1: numbers, if, functions

After lesson 5.

Done	#	Problem	Python you'll use
☐	1	Swap Two Numbers (GfG, Basic)	variables, a temporary variable
☐	2	Last Digit of Number (GfG, Basic)	`% 10`
☐	3	Convert Celsius To Fahrenheit (GfG, Basic)	`*`, `/`, `+`
☐	4	Absolute Value (GfG, Basic)	`if`, or `abs()`
☐	5	Greatest of Three Numbers (GfG, Basic)	`if` / `elif` / `else`, or `max()`
☐	6	Leap Year (GfG, Basic)	`%`, `and`, `or`
☐	7	Function With Return (GfG, Basic)	`def`, `return`
☐	8	Add Two Integers (LeetCode 2235)	`return`, `+`

Stage 2: strings

After lesson 6.

Done	#	Problem	Python you'll use
☐	9	Reverse a String (GfG, Basic)	`[::-1]`
☐	10	Length of String (GfG, Basic)	`len`
☐	11	Convert String to Lower Case (GfG, Basic)	`lower`
☐	12	Lowercase to Uppercase (GfG, Basic)	`upper`
☐	13	Palindrome String (GfG, Basic)	`[::-1]`, `==`

Done	#	Problem	Python you'll use
☐	14	Vowel or Not (GfG, Basic)	`in`
☐	15	Remove Spaces (GfG, Basic)	`replace`
☐	16	Defanging an IP Address (LeetCode 1108)	`replace`
☐	17	To Lower Case (LeetCode 709)	`lower`

Stage 3: lists

After lesson 7.

Done	#	Problem	Python you'll use
☐	18	Largest in Array (GfG, Basic)	`max`
☐	19	Min and Max in Array (GfG, Basic)	`min`, `max`
☐	20	Sum of Array (GfG, Basic)	`sum`
☐	21	Mean or Average of Array (GfG, Basic)	`sum`, `len`, `//`
☐	22	Array End Insert (GfG, Basic)	`append`
☐	23	Concatenation of Array (LeetCode 1929)	`+` on lists
☐	24	Convert the Temperature (LeetCode 2469)	return a list of two numbers

Stage 4: loops

After lesson 8. The biggest stage: most of your practice happens here.

Done	#	Problem	Python you'll use
☐	25	Print GFG n times (GfG, Basic)	`for` + `range`, `print(..., end=" ")`
☐	26	While Loop (GfG, Basic)	`while`
☐	27	Multiplication Table (GfG, Easy)	`for` + `range`, `append`
☐	28	Factorial (GfG, Basic)	`for`, multiply as you go
☐	29	Count Odd and Even (GfG, Basic)	`for`, `%`, two counters
☐	30	Replace all 0's with 5 (GfG, Basic)	`str()`, `replace`, `int()`
☐	31	Even Positioned Characters (GfG, Basic)	`range(0, len(s), 2)`
☐	32	Count Uppercase characters (GfG, Basic)	`for` over a string, `isupper`, counter

Done	#	Problem	Python you'll use
☐	33	Right-Angled Number Triangle Pattern (GfG, Basic)	a loop inside a loop
☐	34	Print Solid Square (GfG, Basic)	a loop inside a loop
☐	35	Running Sum of 1d Array (LeetCode 1480)	running total
☐	36	Shuffle the Array (LeetCode 1470)	`range`, indexes
☐	37	Final Value of Variable After Performing Operations (LeetCode 2011)	`for` over strings, `in`
☐	38	Richest Customer Wealth (LeetCode 1672)	`for` over a list of lists, `sum`, `max`
☐	39	Kids With the Greatest Number of Candies (LeetCode 1431)	`max`, a list of `True` / `False`
☐	40	Maximum Number of Words Found in Sentences (LeetCode 2114)	`split`, `len`, `max`
☐	41	Find Words Containing Character (LeetCode 2942)	`enumerate`, `in`

Stage 5: dictionaries

After lesson 9.

Done	#	Problem	Python you'll use
☐	42	Intro to Dictionary - Python (GfG, Easy)	make a dict, read a key
☐	43	Dictionary in Python - III (GfG, Easy)	add, change and loop over a dict
☐	44	Number of Good Pairs (LeetCode 1512)	count with a dict, or a loop inside a loop

Stage 6: sets and tuples

After lesson 10.

Done	#	Problem	Python you'll use
☐	45	Count Distinct in Array (GfG, Basic)	`set`, `len`
☐	46	Set in Python - II (GfG, Easy)	set operations `&`, `|`, `-`
☐	47	Test if tuple is distinct (GfG, Easy)	`len(set(t)) == len(t)`
☐	48	Search in Python Tuple (GfG, Easy)	tuples, `index`, `in`
☐	49	Jewels and Stones (LeetCode 771)	`set`, `in`
☐	50	Count the Number of Consistent Strings (LeetCode 1684)	`set`, `in`